

Predictive Modeling of Citi Bike User Types

May 2026

Introduction and Feature Engineering

Data has been sourced from the Citi Bik system data site.

The data that has been utilized for this classification project is from March 2026.

Pre-feature engineering the following independent variables were available.

- Rideable type: electric or non-electric bike
- Start time and end time in YYYY-DD-MM, 24:00:00:000 format
- Start station name, start station id, end station name, end station id
- Start latitude, Start longitude, end latitude, and end longitude
- Member or Casual rider

In order to perform a random forest classification analysis, features were engineered from existing variables, binned and then encoded.

Duration: A duration variable was created based off the start and end time. The pandas Datetime function was used to ensure proper data handling. Data was then formatted into MM: SS.

$$\text{Duration} = \text{End Time} - \text{Start Time}$$

Duration was mapped as follows (in minutes)

$$\text{Very short } 0 < x \leq 10$$

$$\text{Short } 10 < x \leq 15$$

$$\text{Medium } 15 < x \leq 30$$

$$\text{Long } 30 < x \leq 60$$

$$\text{Very Long } 60 < x \leq 120$$

$$\text{Misdocked? } x > 120$$

Start and End Borough: Utilizing a Borough Boundaries data set, obtained from New York open-source data, the start and end station were mapped to the associated borough based off of the latitude and longitude.

Data was then binned based on the time of day. Early Morning from midnight to 5:59 AM, morning from 6 AM to 11:59 AM, Afternoon from 12:00 PM to 3:59 PM, evening from 4:00 PM to 5:59 PM, Night from 6:00PM to 11:50 PM.

Then a trip type column was created that mapped whether the rider started and ended in the same borough – Interborough. If the rider started and ended in a different borough – out of borough.

Then based off of start *time* a weekend or weekday variable was encoded, 0 for weekday and 1 for weekend.

The data frame was then cleaned, and irrelevant columns were dropped prior to any analysis. Once cleaning and feature engineering was complete the following variables were remaining:

- Ride type
- Member or casual rider
- Rider time in minutes
- The duration of the ride time: very short, short, medium, long, very long, or misdocked?
- Start borough and end borough
- Start time of day and end time of day: Early Morning, morning, afternoon, evening, night
- Trip type – interborough or out of borough
- Weekend of weekday.

Objective

The objective of this analysis is to determine if a Citi Bike user is a member or casual rider, based off of ride duration, weekend rider or weekday rider, interborough or out of borough trip, peddle bike or e-bike, and the starting and ending time of the ride.

Method – Random Forest

Three Random Forest models were utilized in this analysis: unbalanced, the base line model. Raw data was used without any sort of splitting or organizing.

A balanced model was then used, keeping the original population size of the data but applying a penalty for misclassification of the minority class, casual rider.

A 50/50 split was then utilized to handle the glaring class imbalance. A smaller dataset was created from the original population, utilizing a near 50/50 spit for member and casual riders.

For all models a split of 80% training and 20% testing was utilized during the evaluation.

Method – Logistic Regression

After the Random Forest models were constructed, a logistic regression was performed on the 50/50 dataset. The goal was to reinforce the information provided by the 50/50 random forest model and to see whether additional insights could be gained.

Analysis and Interpretation

For the random forest models, standard evaluation metrics were used to gauge each model and the insight each provided: Accuracy, precision, recall, and F-1 score.

Metrics are defined as follows in the context of the objective

Accuracy: The proportion of correct predictions, true positive and true negative, out of the total number of predictions made on the dataset. How often does the model correctly identify a causal or member rider.

$$Accuracy = \frac{\text{Total Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

Precision: The ratio of correctly predicted positive observation to the total number of predicted positive observations. Of all users the model labeled as member (casual), what percentage were actually members (casual).

$$Precision = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Recall: Measure's the model's ability to find all relevant instances within a specific class. Of all the members (casual) riders predicted, how many did the model get correct?

$$Recall = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

F-1 score: The harmonic mean of precision and recall. The harmonic means penalizes extreme values ensuring that a high score can only be achieved if both precision and recall are independently strong which gives a more accurate measure of model stability than just accuracy alone.

The f1 score provides a metric between 0 and 1, the higher the number implies better performance as it reflects a higher precision and recall value. Ensures that the model isn't just guessing one class while ignoring the other.

$$F1\ Score = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Model Results

Unbalanced model

	Precision	Recall	f1-score	Support
Casual	0.61	0.04	0.08	81,171
Member	0.87	1.00	0.93	507,980
Accuracy			0.86	589,151

While the model scored 86% within the accuracy metric, the high score is due to the difference in support size, 81.1k casual riders versus 507.9k member riders. The recall metric for both casual rider and member rider are 0.04 and 1.0, respectively. This shows that the model is guessing member riders for every test point, missing 96% of casual riders.

The low f1 score for casual riders reinforces that the model is essentially useless for predicting whether or not a user is casual or a member because the model will always return member.

In summary the model is able predict, with 86% accuracy, by just guessing member rider. So, the model is not actually learning anything about the data.

As a way to remediate the class imbalance issue a balanced model was introduced.

Balanced Model

	Precision	Recall	f1-score	Support
Casual	0.22	0.59	0.32	81,171
Member	0.91	0.67	0.77	507,980
Accuracy			0.66	589,151

The balance model punishes for misclassified results, which can be seen in a decrease in accuracy with a rise in precision for casual riders. The model is now 66% accurate with a casual rider recall score of 22%, up from just 4%. Although precision decreased to 22% from 61%, indicates a high false positive rate, predicting a casual rider when in fact the rider is a member.

In the balanced model the minority class, casual rider, is prioritized, meaning the model expands its decision boundary in hopes of capturing more casual riders. But due to the extreme class imbalance the decision boundary naturally captures a massive amount of members who share traits with casual riders, these represent the false positives and are the driving reason precision decreased.

The balanced model targets more broadly rather than accurately. More casual riders are being identified but 78% of member riders are being classified as casual.

50/50 Model

	Precision	Recall	f1-score	Support
Casual	0.63	0.61	0.62	81,412
Member	0.62	0.65	0.63	81,606
Accuracy			0.63	163,018

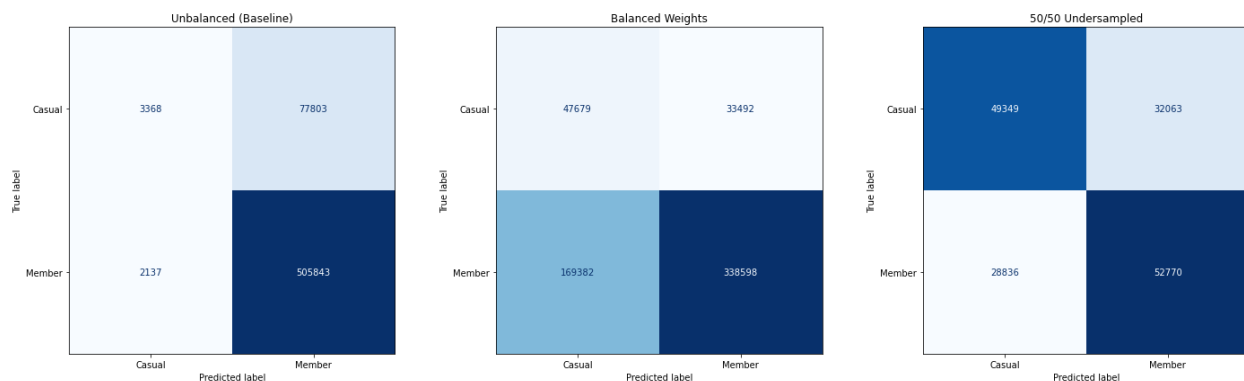
The 50/50 model splits members and casual riders into two almost equal groups, down sampling. By down sampling the majority class is removed lowering the chance of false positives.

The results returned are almost symmetric.

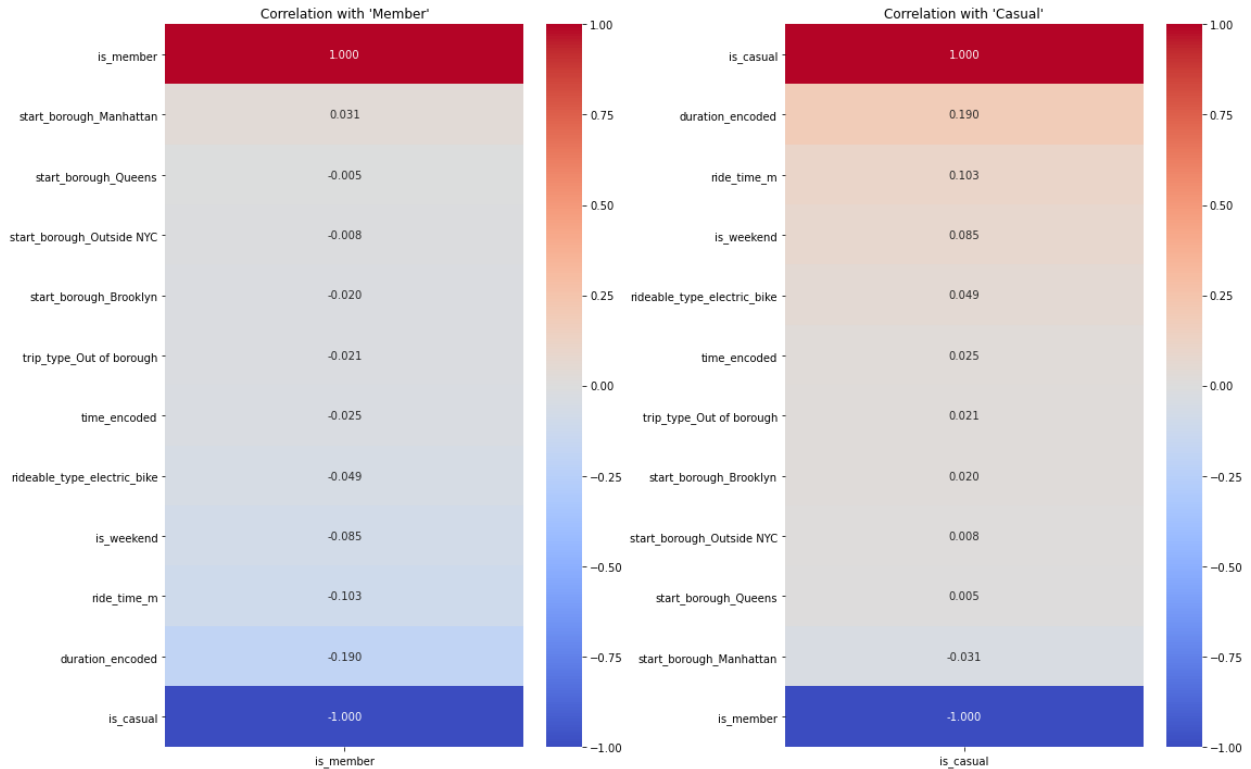
The lower f1 score for member and the higher f1 score for casual represents the effects of balancing the population. While the f1 score is generally lower than the unbalanced baseline model, it is a more sound reflection of casual versus member riding behavior. For the casual rider a 0.62 f1 score is considered stable and provides a good middle ground of stable discovery while minimizing the false positive rate.

Overall accuracy reduced to 63%, which was expected. While precision and recall both rose for the casual rider. Which shows that the 50/50 model is much more actionable and reliable for classifying casual or member riders than the balanced or unbalanced model did.

Confusion Matrix for the three Random Forest Models



Coefficient Signals for Causal and Member – Random Forest Models



Logistic Regression – 50/50 Model

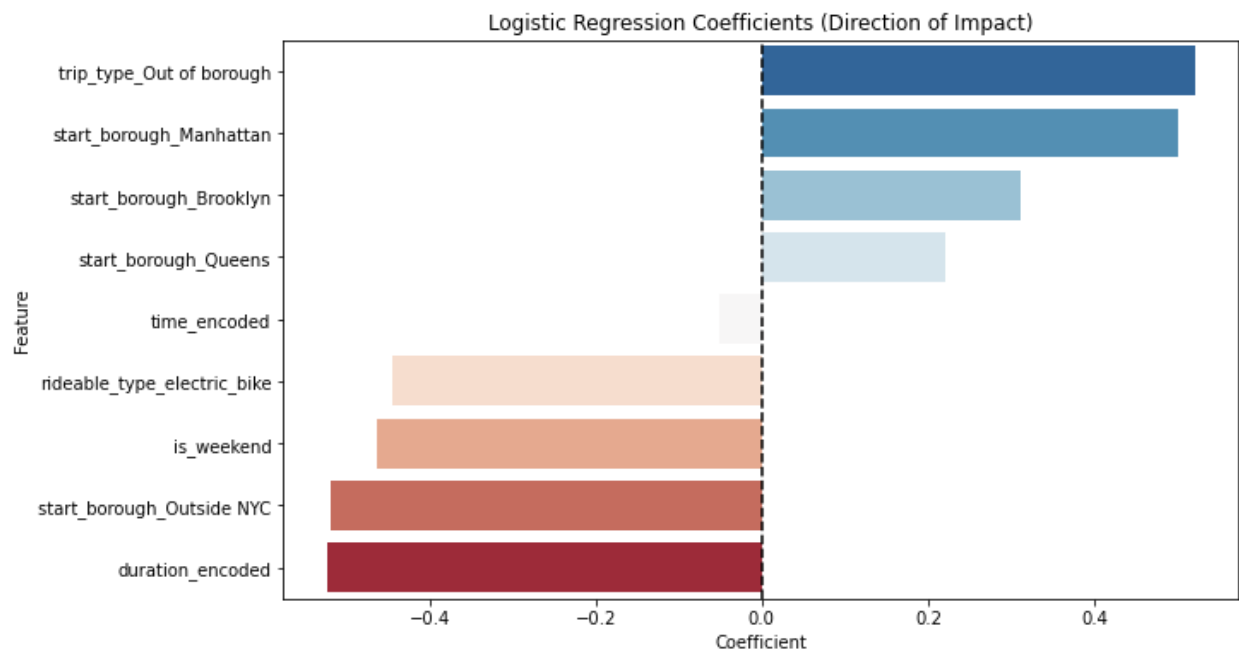
After the random forest models were run and evaluated a logistic regression model was run on the model with the most stable evaluation metrics, 50/50 model, the results were then evaluated using the same KPIs.

Logistic Regression				
	Precision	Recall	f1-score	Support
Casual	0.62	0.58	0.60	81,412
Member	0.65	0.65	0.63	81,606
Accuracy			0.61	163,018

The logistic regression returns similar results to the 50/50 model. The random forest models are considered nonlinear while the logistic regression model is considered linear. The f1 results, for casual riders, were 0.6 logistic regression and 0.62 random forest, an agreement signal. The fact that two different types of models returned similar f1 metrics gives evidence that the data is both fundamental and linear.

The model finds clear signals which indicate casual rider. An agreement signal implies that the random forest is not “memorizing” the data. And since logistic regression is a high bias and low variance model it is constrained by linear decision bounds which prevents overfitting to noise, akin to data memorization. So the 5050 model is deemed useful for the analysis of March 2026 rider behavior.

Coefficient Signals for Causal and Member – Logistic regression



The coefficients chart above shows the impact the features have on the target variable for the logistic regression. Zero is the neutral point everything to the right of the model, in blue, points towards a member rider. Everything to the left of zero, in red, points towards a casual rider. Two of the strongest signals for casual riders are duration encoded and is weekend, meaning that longer riders and weekends tend to be more geared towards casual rides rather than member rides.

Additional Finding

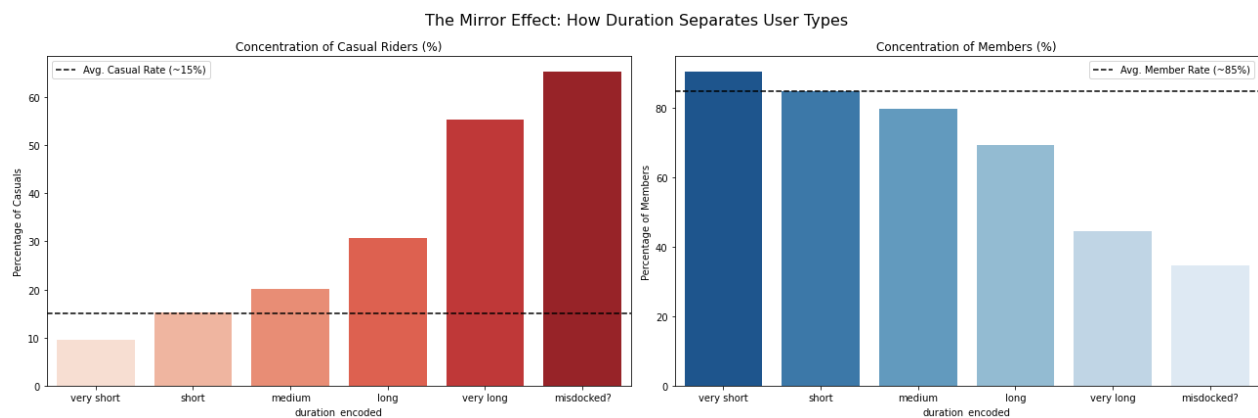
Below is a chart that describes the top three most important features for each model. The random forest is based off the contribution score, a number between 0 and 1 that tells what percentage of the model’s decision comes from that feature. Note that all the contributions score, from all features of the random forest model, will sum to 1.

The logistic regression model is evaluated off of weight scores, which represents the magnitude of the coefficients, the strength of a relationship between a feature and the log-odds of a rider being a member or casual rider. The weights are not in proportion and will not sum to one.

Random Forest Top Features – Contribution	
Duration encoded	0.602
is weekend	0.117
time encoded	0.103
Logistic Regression – Weights	
Duration encoded	0.523
Trip type (Out of borough)	0.521
Start borough (Outside of NYC)	0.519

Interpreting the strongest coefficient for each model, duration encoded, means that within the Random Forest model 60% of the model’s decision making comes from the duration encoded variable.

Within the logistic regression model the duration encoded variable means that for every step up in duration the model significantly shifts towards predicting a casual rider. It is important to note that within the logistic regression model trip type and start borough also provide just as much decision-making ability as duration.



Member versus Casual Rider and Duration can be seen below.

Duration Category	Casual Rider (%)	Annual Member (%)	Total (%)
Very Short (<5-10m)	9.51%	90.49%	100%
Short	15.22%	84.78%	100%
Medium	20.22%	79.78%	100%
Long	30.66%	69.34%	100%
Very Long	55.35%	44.65%	100%
Misdocked?	65.24%	34.76%	100%

Summary

After engineering, cleaning, and analyzing the data using a random forest model and a logistic regression model the big behavioral findings are that over 90% of trips under 5 minutes are from member riders, as trips extend past 30 minutes the concentration of casual riders increases exponentially, and casual riders are more likely to ride on the weekend than member riders.

All of these points highlight that for March 2026 the casual rider is more likely to ride for leisure and enjoyment whereas the member rider is more likely to be a commuter looking to get from point A to point B.

Considerations

Some project specific considerations are that this data is only from March 2026. While the data is very robust, time of year and seasonality may have measurable impacts on the activity and behavior of a casual rider versus a member rider. The current model potentially underrepresents leisure behavior that occurs in high numbers during peak summer months. Also, an e-bike analysis was not conducted, a 15-minute ride on an ebike is much further than 15 minutes on a peddle assisted bike. Which means that an analysis could be conducted on the behavior of rider status and bike type and the impact on duration.

Sources

Citi Bike System Data (NYC Open Data Portal) - New York City Department of Transportation. "Citi Bike System Data." *NYC Open Data*, 2026, data.cityofnewyork.us/dataset/Citi-Bike-System-Data/vsnr-94wk/about_data. Accessed 11 May 2026.

Borough Boundaries (NYC Open Data Portal) - New York City Department of City Planning. "Borough Boundaries." *NYC Open Data*, 2026, data.cityofnewyork.us/City-Government/Borough-Boundaries/gthc-hcne/about_data. Accessed 11 May 2026.

Citi Bike Official System Data Page - Lyft Bikes and Scooters, LLC. "System Data." *Citi Bike NYC*, 2026, citibikenyc.com/system-data. Accessed 11 May 2026.

Géron, Aurélien. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*. 2nd ed., O'Reilly Media, 2019.

Gemini 3 Flash, 11 May 2026